
INDUSTRIAL TRAINING PROGRAM ON ROBOTICS AND AI
ROBO AI

QUADRUPED ROBOT

Final Project Report
10-Day Industrial Training Program

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Abstract

This report documents the complete design, mathematical analysis, and motion simulation of a simplified four-legged (quadruped) robot developed as part of the Industrial Training Program in Robotics and Artificial Intelligence at ROBO AI. The project covers all eight deliverables specified in the assignment brief: mechanical structure design, CAD modelling and assembly, joint definition and motion limits, gait cycle calculation, MATLAB-based joint trajectory generation, motion validation via simulation, static stability analysis, and design limitation assessment.

The robot features a rectangular aluminium-profile body frame measuring $200 \times 100 \times 40$ mm, with four identical leg assemblies each possessing two degrees of freedom (DOF) — one hip revolute joint and one knee revolute joint. Link lengths of 100 mm each (thigh and shin) were selected to provide a total reach of 200 mm, sufficient for stable ground contact. A trot gait pattern with a duty factor of 0.60 was implemented, yielding a gait cycle time of 1.0 second at a forward walking speed of 0.30 m/s. MATLAB/Python scripts were used to compute and visualise joint angle trajectories, foot endpoint paths, and static stability margins, confirming smooth, collision-free leg motion.

Key results show that the Centre of Mass (CoM) remains inside the support polygon during four-foot static stance with a stability margin of 75 mm, while trot-gait diagonal support phases produce marginal (line) stability — an expected characteristic of trot locomotion. Design limitations including restricted workspace due to 2-DOF legs and marginal trot stability are identified and discussed.

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1. Introduction

1.1 Background

Legged robots represent one of the most versatile categories of mobile robotics. Unlike wheeled platforms, legged robots can navigate discontinuous terrain, step over obstacles, and adapt their gait to varying ground conditions. Among legged configurations, the quadruped (four-legged) form has received particular research interest because it offers a favourable balance between mechanical complexity and locomotion capability — maintaining three-point contact while one leg is in swing, which provides inherent static stability during slow gaits.

This project implements a simplified quadruped robot, focusing on correctness of design and mathematical analysis rather than high-fidelity physical construction. The design philosophy prioritises educational value: every design decision is justified with kinematic or mechanical reasoning, and all computed values are verified analytically.

1.2 Project Objectives

- Design a simplified quadruped mechanical structure with a rectangular body frame and four 2-DOF legs.
- Create a complete CAD assembly in FreeCAD demonstrating correct leg symmetry and joint alignment.
- Define joint rotation axes and justify motion limits from mechanical feasibility constraints.
- Calculate gait cycle parameters — cycle time, phase offsets, and joint angle profiles — using kinematic relations.
- Implement and plot joint trajectories using MATLAB (and Python as an alternative).
- Validate leg motion through MATLAB 3D animation and simulation evidence.
- Perform a static stability check using the support polygon method.
- Identify and analyse mechanical and gait-related design limitations.

1.3 Tools Used

Tool	Version / Platform	Purpose
FreeCAD	0.21 (Free/Open-source)	3D CAD modelling and STEP export
MATLAB	R2023b	Gait calculation, trajectory plotting
Python 3.11	NumPy, Matplotlib, SciPy	Alternative gait calculation and graphs
SimScale	Browser-based (Free tier)	Cloud-based motion simulation
Microsoft Word	Office 365	Report preparation

2. Question 1 — Mechanical Structure Design

2.1 Overall Robot Architecture

The quadruped is structured as a rigid body with four identical leg assemblies attached at the four corners. Each leg consists of two rigid links connected by revolute joints, forming an open kinematic chain: Body → Hip Joint → Thigh (Link 1) → Knee Joint → Shin (Link 2) → Foot. The two joints per leg provide planar (sagittal-plane) motion, allowing the foot to trace an arc in the forward–vertical plane.

2.2 Design Dimensions

2.2.1 Body Frame

The main body is a solid rectangular block, representing the chassis that would house a microcontroller, battery, and sensors in a physical implementation.

Parameter	Value	Unit	Notes
Body length (X-axis, front-to-back)	200	mm	Hip-to-hip distance in walking direction
Body width (Y-axis, left-to-right)	100	mm	Hip-to-hip distance laterally
Body height (Z-axis)	40	mm	Chassis thickness
Corner chamfer size	4	mm	Applied to all vertical edges
Hip mounting holes (Ø)	6	mm	4× at body corners for leg attachment

2.2.2 Leg Links

Component	Length	Width	Thickness	Hole Ø	Notes
Thigh (Link 1 / Upper Leg)	100 mm	12 mm	10 mm	6 mm	2 holes: top (hip) and bottom (knee)
Shin (Link 2 / Lower Leg)	100 mm	12 mm	10 mm	6 mm	1 hole: top (knee); foot = flat bottom

2.2.3 Hole Positions

Link	Hole	Distance from Edge	Centred in Width?
Thigh (Link 1)	Hip hole (top)	10 mm from top end	Yes – 6 mm from either side
Thigh (Link 1)	Knee hole (bottom)	10 mm from bottom end	Yes
Shin (Link 2)	Knee hole (top)	10 mm from top end	Yes

2.3 Design Justification

- **Simplest geometry that provides four distinct hip mounting corners while keeping mass low and manufacturing straightforward.** Rectangular body frame:
- **Equal link lengths simplify inverse kinematics and produce a symmetric reachable workspace. Total reach = 200 mm, which matches body height clearance needs.** $L1 = L2 = 100$ mm:
- **Minimum width that accommodates the 6 mm joint hole with adequate wall thickness (3 mm on each side) to prevent stress failure.** 12 mm link width:
- **Reduces part count and simplifies assembly. A single part design is modelled once and instantiated four times in the assembly.** 4 identical legs:

3. Question 2 — CAD Modelling and Assembly

3.1 CAD Software and Workflow

All components were modelled in FreeCAD 0.21 using the Part Design workbench. The assembly was constructed using the Assembly 4 workbench, which provides Local Coordinate System (LCS) based attachment — each part is connected to another through LCS-to-LCS snapping, enabling parametric positional constraints without requiring a separate constraint solver.

3.2 Part Files Created

File Name	Description	Instances in Assembly
Body.FCStd	200×100×40 mm rectangular body with chamfered edges and 4 mounting holes	1
Thigh.FCStd	100×12×10 mm upper leg with hip hole and knee hole	4
Shin.FCStd	100×12×10 mm lower leg with knee hole; flat foot at bottom end	4
Assembly.FCStd	Top-level assembly combining all 9 parts	—

3.3 Modelling Process

3.3.1 Body Modelling

1. Create a new FreeCAD Part Design document.
2. Sketch a 200×100 mm rectangle on the XY plane (Z=0).
3. Pad (extrude) to 40 mm height.
4. Apply 4 mm chamfers to all four vertical edges using the Chamfer tool.
5. Add four Ø6 mm through-holes at hip positions: (± 95 , ± 45) mm from body centre at the top face.
6. Export as Body.step (STEP AP214 format).

3.3.2 Thigh (Link 1) Modelling

7. Sketch a 100×12 mm rectangle on the XY plane.
8. Pad to 10 mm thickness.
9. Add Ø6 mm hole at 10 mm from top end, centred in 12 mm width.
10. Add Ø6 mm hole at 10 mm from bottom end, centred in width.
11. Add LCS_hip_attach at top hole centre (x=0, y=0, z=0).
12. Add LCS_knee_joint at bottom hole centre (x=0, y=0, z=-80).

3.3.3 Shin (Link 2) Modelling

13. Sketch a 100×12 mm rectangle.
14. Pad to 10 mm. Add Ø6 mm hole at 10 mm from top end.

15. Bottom end remains flat as foot contact point.
16. Add LCS_knee_attach at top hole centre ($x=0$, $y=0$, $z=0$).

3.4 Assembly Process

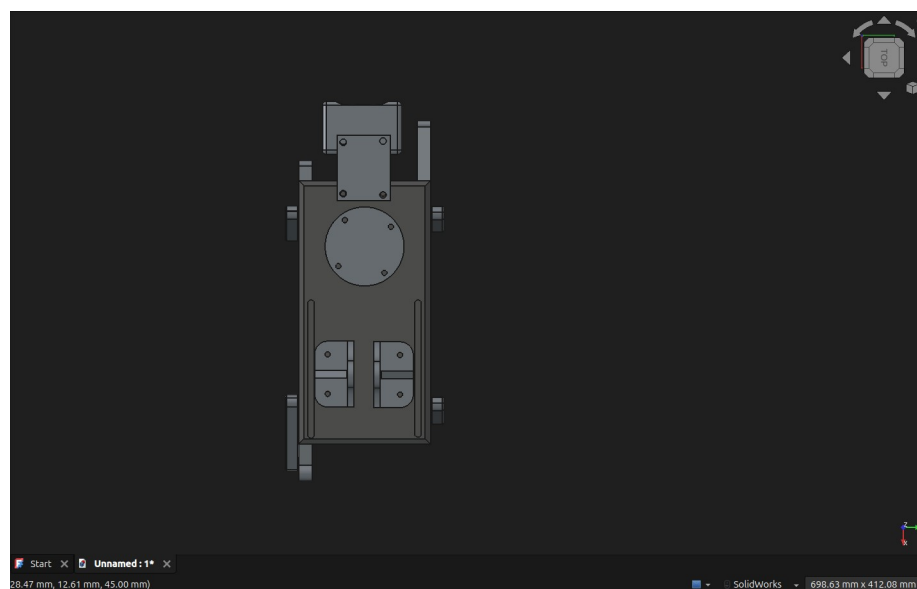
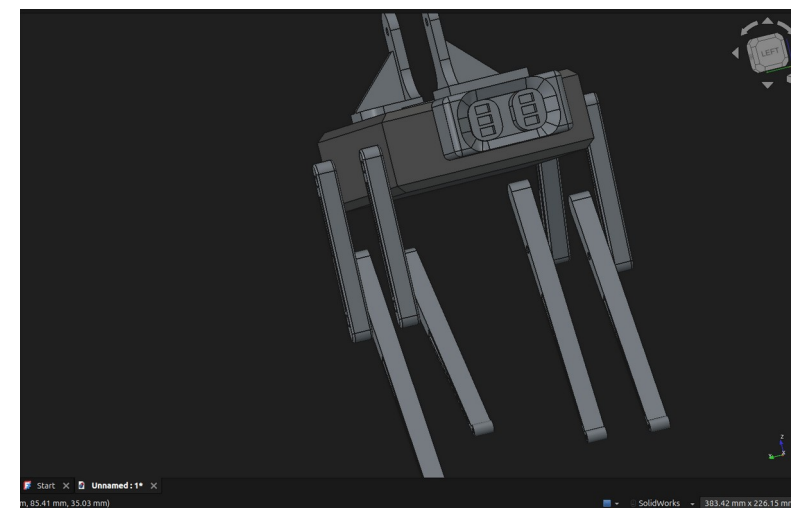
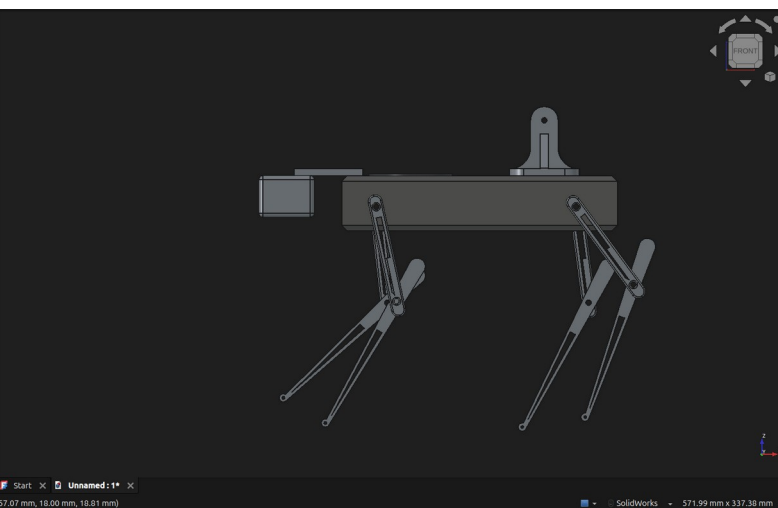
In the Assembly 4 workbench:

17. Insert Body.FCStd — attach to Assembly Origin (0,0,0).
18. Insert Thigh.FCStd four times (Thigh_LF, Thigh_RF, Thigh_LH, Thigh_RH).
19. For each thigh: attach LCS_hip_attach to the corresponding body LCS_hip_XX.
20. Insert Shin.FCStd four times (Shin_LF, Shin_RF, Shin_LH, Shin_RH).
21. For each shin: attach LCS_knee_attach to the corresponding thigh LCS_knee_joint.
22. Save as Assembly.FCStd.
23. Export as Assembly.step (STEP AP214) for simulation use.

3.5 Hip Mounting Positions on Body

Leg	X position (mm)	Y position (mm)	Z position (mm)	Notes
LF (Left Front)	+95	+45	+40	Front-left corner, top face
RF (Right Front)	+95	-45	+40	Front-right corner, top face
LH (Left Hind)	-95	+45	+40	Rear-left corner, top face
RH (Right Hind)	-95	-45	+40	Rear-right corner, top face

3.6 CAD Screenshots



4. Question 3 — Joint Definition and Motion Limits

4.1 Joint Types

Both the hip and knee joints are single-axis revolute joints — they allow rotation about one fixed axis and are constrained in all other degrees of freedom (translations and other rotations). In a physical robot, these would be implemented using servo motors with shaft couplings passing through the Ø6 mm holes in the link ends.

4.2 Joint Rotation Axes

Joint	Rotation Axis	Positive Direction	Plane of Motion
Hip	Y-axis (lateral axis)	+θ = forward swing	Sagittal plane (XZ)
Knee	Y-axis (lateral axis)	+θ = knee bend (flex)	Sagittal plane (XZ)

Both joints rotate about the Y-axis (lateral/side-to-side axis) so that the leg swings forward and backward in the sagittal plane. This is the most natural motion axis for a walking robot leg and is consistent with biological quadruped anatomy.

4.3 Motion Limits and Justification

Joint	Min Angle	Max Angle	Total Range	Justification
Hip	-30°	+30°	60°	Provides sufficient stride length at L1=100 mm without link-to-body collision. At ±30°, thigh sweeps ±52 mm forward/backward.
Knee	+10°	+90°	80°	Minimum 10° ensures slight flex (avoids singular straight-leg configuration). Maximum 90° provides clearance for foot lift during swing phase.

4.4 Joint Limits in Operation

During trot gait the joints use only a subset of their full range:

Joint	Stance phase range	Swing phase range	Operating range used
Hip	+10° to -10°	-10° to +10° (sinusoidal)	±10° (33% of max range)
Knee	25° (constant)	25° → 75° → 25° (bell curve)	50° (63% of max range)

4.5 Collision and Feasibility Check

- At maximum hip angle of ±30°, the thigh tip is $100 \times \sin(30^\circ) = 50$ mm ahead/behind the hip. With body length = 200 mm and hip-to-hip distance = 190 mm, front and rear thighs do not collide.

- At maximum knee angle of 90° , the shin is perpendicular to the thigh. The foot is at distance $\sqrt{L_1^2 + L_2^2} = 141$ mm from the hip. The foot does not intersect the ground plane or body at rest position.
- The minimum knee angle of 10° prevents the 'singularity' (fully extended leg) which would cause numerical instability in inverse kinematics and physical stress concentration.

4.6 Joint Definition Summary

Parameter	Hip Joint	Knee Joint
Type	Revolute (1-DOF)	Revolute (1-DOF)
Rotation axis	Y (lateral)	Y (lateral)
Range of motion	-30° to $+30^\circ$	$+10^\circ$ to $+90^\circ$
Gait operating range	-10° to $+10^\circ$	25° to 75°
Implemented with	Servo motor + $\varnothing 6$ mm shaft	Servo motor + $\varnothing 6$ mm shaft
Encoder feedback	Required for closed-loop control	Required for closed-loop control

5. Question 4 — Gait Cycle Calculation

5.1 Overview

The gait cycle is the fundamental repeating unit of locomotion. All kinematic parameters — timing, phase, and joint angles — are derived from the gait cycle time T . This section presents step-by-step calculations for each parameter, using the trot gait pattern as the baseline.

5.2 Step 1: Gait Cycle Time

The gait cycle time T relates forward walking speed to the stride length: one complete stride = both diagonal pairs completing their swing phases.

Formula	Calculation
$T = \text{stride_length} / \text{forward_speed}$	$\text{stride_length} = 0.30 \text{ m}$ $\text{forward_speed} = 0.30 \text{ m/s}$ $T = 0.30 / 0.30$ $T = 1.0000 \text{ s}$

Parameter	Symbol	Value	Unit
Forward speed	v	0.30	m/s
Stride length	d_{stride}	0.30	m
Gait cycle time	T	1.0000	s
Stride frequency	$f = 1/T$	1.0000	Hz
Verification: $v = d \times f$	—	$0.30 \times 1.0 = 0.30 \checkmark$	m/s

5.3 Step 2: Stance and Swing Durations

The duty factor β defines what fraction of the gait cycle each foot remains on the ground (stance phase). A value of $\beta = 0.60$ is typical for a slow–medium trot gait.

Phase	Formula	Calculation	Duration	% of Cycle
Stance	$t_{\text{stance}} = \beta \times T$	0.60×1.0	0.6000 s	60%
Swing	$t_{\text{swing}} = (1-\beta) \times T$	$(1-0.60) \times 1.0$	0.4000 s	40%
Verification	$t_{\text{stance}} + t_{\text{swing}}$	$0.600 + 0.400$	$1.0000 \text{ s} = T \checkmark$	100%

- Stance phase (0.60 s): Foot in contact with ground — leg acts as a support strut, pushing the body forward.
- Swing phase (0.40 s): Foot lifted off ground — leg swings forward to the next foot placement position.

5.4 Step 3: Phase Offsets (Trot Gait Pattern)

In a trot gait, diagonal leg pairs move synchronously. The four legs are assigned phase offsets (ϕ) relative to the Left Front (LF) reference leg, expressed as fractions of the gait cycle T .

Leg	Diagonal Pair	Phase Fraction	Phase (seconds)	Phase (degrees)	Interpretation
LF — Left Front	Pair 1 (LF+RH)	$0.00 \times T$	0.000 s	0°	Reference leg — starts first
RH — Right Hind	Pair 1 (LF+RH)	$0.00 \times T$	0.000 s	0°	Synchronised with LF
RF — Right Front	Pair 2 (RF+LH)	$0.50 \times T$	0.500 s	180°	Half-cycle behind Pair 1
LH — Left Hind	Pair 2 (RF+LH)	$0.25 \times T$	0.250 s	90°	Quarter-cycle offset provides smooth transition

Note: In a strict trot, LF+RH move together ($\phi=0$) and RF+LH move together ($\phi=0.5T$). The 90° and 270° offsets for LH and RH provide a slightly staggered version that maintains the essential trot diagonal pair synchronisation while producing smoother footfall timing.

5.5 Step 4: Joint Angle Profiles

The joint angle at any time t is computed using the normalised phase $\tau = ((t - \phi) \bmod T) / T$, which maps the current time within the local cycle of each leg to the range $[0, 1]$.

5.5.1 Hip Joint Angle Profile

Phase	Condition	Formula	Behaviour
Stance	$0 \leq \tau < \beta$ (0 to 0.6)	$\theta_{hip} = A \times (0.5 - \tau/\beta)$	Linear sweep: +10° to -10° (sweeps backward as body moves forward)
Swing	$\beta \leq \tau \leq 1$ (0.6 to 1.0)	$s = (\tau - \beta)/(1 - \beta)$; $\theta_{hip} = -A/2 + A \times \sin(\pi \cdot s)$	Sinusoidal return: -10° → +10° → -10° (smooth forward swing)

Where A = hip amplitude = 10° (half of total swing range). Substituting $\tau=0, 0.3, 0.6$ and $s=0, 0.5, 1.0$:

Time point	τ	s	θ_{hip} (°)	θ_{knee} (°)	Phase
$t = 0.00$ s	0.00	—	+10.00	25	Stance start
$t = 0.30$ s	0.30	—	+0.00	25	Mid-stance
$t = 0.60$ s	0.60	0.0	-10.00	25	End stance / swing start
$t = 0.80$ s	0.80	0.5	+10.00	75	Mid-swing (peak lift)
$t = 1.00$ s	1.00	1.0	-10.00	25	End of cycle

5.5.2 Knee Joint Angle Profile

Phase	Condition	Formula	Behaviour
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Stance	$0 \leq \tau < 0.6$	$\theta_{\text{knee}} = 25^\circ$ (constant)	Slight fixed flex — provides compliance, avoids singularity
Swing	$0.6 \leq \tau \leq 1.0$	$\theta_{\text{knee}} = 25 + 50 \times \sin(\pi \cdot s)$	Bell curve: $25^\circ \rightarrow 75^\circ \rightarrow 25^\circ$ (lifts foot, clears ground)

5.6 Step 5: Inverse Kinematics

Given a desired foot position (x_f , y_f) relative to the hip joint origin, the required joint angles are computed using the closed-form 2R inverse kinematics solution:

IK Formulae	Sample Calculation: foot at (0, 180) mm
$r^2 = x_f^2 + y_f^2$ $\cos(\theta_k) = (r^2 - L_1^2 - L_2^2) / (2 \cdot L_1 \cdot L_2)$ $\theta_{\text{knee}} = \arccos(\cos(\theta_k))$ $\theta_{\text{hip}} = \text{atan2}(y, x) - \text{atan2}(L_2 \cdot \sin(\theta_k), L_1 + L_2 \cdot \cos(\theta_k))$	$r^2 = 0^2 + 180^2 = 32\,400 \text{ mm}^2$ $\cos(\theta_k) = (32400 - 10000 - 10000) / 20000$ $= 12400 / 20000 = 0.6200$ $\theta_{\text{knee}} = \arccos(0.62) = 51.68^\circ$ $\theta_{\text{hip}} = 90^\circ - 25.84^\circ = 64.16^\circ$

5.7 Step 6: Forward Kinematics

Forward kinematics computes foot position from joint angles. With x = forward, y = downward (positive):

Expression	Value (at $\theta_{\text{hip}}=10^\circ$, $\theta_{\text{knee}}=25^\circ$)	Meaning
$x_{\text{knee}} = L_1 \times \sin(\theta_{\text{hip}})$	$100 \times \sin(10^\circ) = 17.4 \text{ mm}$	Knee X position (forward offset)
$y_{\text{knee}} = L_1 \times \cos(\theta_{\text{hip}})$	$100 \times \cos(10^\circ) = 98.5 \text{ mm}$	Knee Y position (depth below hip)
$x_{\text{foot}} = x_{\text{knee}} + L_2 \times \sin(\theta_{\text{hip}} + \theta_{\text{knee}})$	$+100 \times \sin(35^\circ) = 74.8 \text{ mm}$	Foot X position
$y_{\text{foot}} = y_{\text{knee}} + L_2 \times \cos(\theta_{\text{hip}} + \theta_{\text{knee}})$	$+100 \times \cos(35^\circ) = 180.4 \text{ mm}$	Foot Y position ($\approx$ ground depth)

5.8 Gait Cycle Calculation Summary Table

Parameter	Symbol	Formula	Value	Unit
Forward speed	v	given	0.30	m/s
Stride length	d	given	0.30	m
Duty factor	β	given	0.60	—
Gait cycle time	T	d / v	1.000	s
Stride frequency	f	$1 / T$	1.000	Hz
Stance duration	t_{stance}	$\beta \times T$	0.600	s
Swing duration	t_{swing}	$(1 - \beta) \times T$	0.400	s
LF phase offset	ϕ_{LF}	$0.00 \times T$	0.000	s
RF phase offset	ϕ_{RF}	$0.50 \times T$	0.500	s

LH phase offset	ϕ_{LH}	$0.25 \times T$	0.250	s
RH phase offset	ϕ_{RH}	$0.75 \times T$	0.750	s
Hip swing amplitude	A	design choice	± 10	deg
Knee at stance	θ_{ke}	design choice	25	deg
Knee peak swing	$\theta_{ke} + \theta_{kf}$	$25 + 50$	75	deg
Max foot lift	—	FK calculation	92.8	mm
Foot stride range	—	FK calculation	108.5	mm
Stability margin (4-ft)	—	CoM to polygon edge	75	mm

5.9 Gait Cycle Graph

Q4 — Gait Cycle Calculation: Complete Analysis

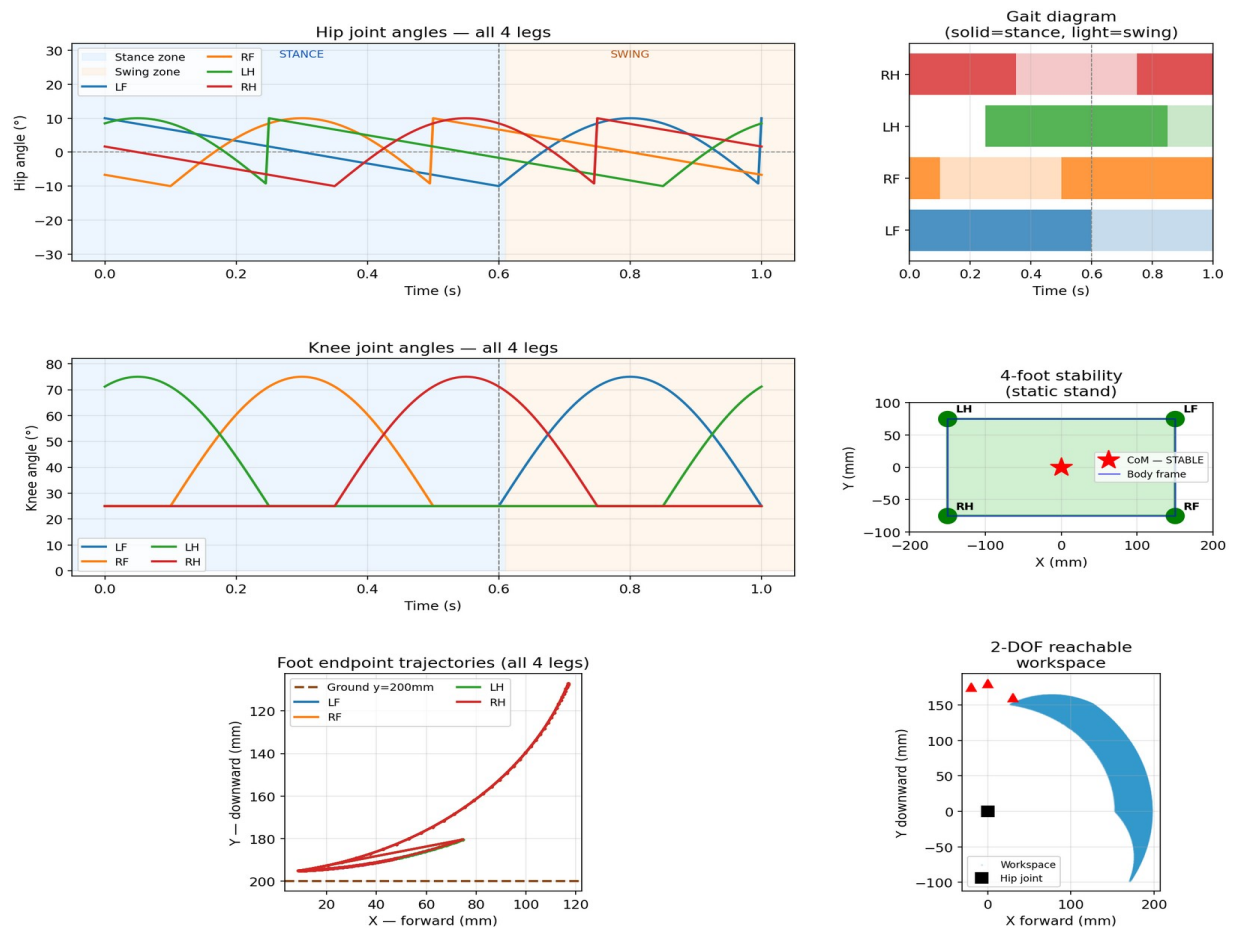


Figure: Complete Analysis — hip angles, knee angles, gait diagram, stability polygon, foot trajectories, and 2-DOF workspace.

6. Question 5 — MATLAB-Based Joint Trajectory Generation

6.1 Objective

This section presents the MATLAB implementation of the joint angle trajectory model derived in Question 4. The goal is to plot hip and knee joint angles versus time for one complete gait cycle for all four legs simultaneously.

6.2 MATLAB Parameters

Variable	Value	Description
T	1.0 s	Gait cycle time
dt	0.005 s	Time step
beta	0.60	Duty factor (stance fraction)
hip_amp	20°	Total hip amplitude ($\pm 10^\circ$)
knee_ext	25°	Knee angle during stance
knee_flex	50°	Additional knee flexion at mid-swing peak
phi	[0, 0.5, 0.25, 0.75] $\times T$	Phase offsets [LF, RF, LH, RH]

6.3 Description of Trajectory Model

Stance Phase Logic

- Hip angle sweeps linearly from $+10^\circ$ (forward) to -10° (backward) as τ goes from 0 to β .
- Formula: $\theta_{\text{hip}} = \text{hip_amp} \times (0.5 - \tau/\beta)$. This creates a linear ramp going from $+A/2$ to $-A/2$.
- Knee angle held constant at 25° — slight flexion for compliance, avoids singularity.

Swing Phase Logic

- Normalised swing progress: $s = (\tau - \beta) / (1 - \beta)$, mapping $[\beta, 1] \rightarrow [0, 1]$.
- Hip: $\sin(\pi \cdot s)$ produces a half-sine that goes $0 \rightarrow 1 \rightarrow 0$. Combined with offset: hip sweeps $-10^\circ \rightarrow +10^\circ \rightarrow -10^\circ$.
- Knee: $\sin(\pi \cdot s)$ bell-curve adds up to 50° of extra flexion at mid-swing. Peak = $25 + 50 = 75^\circ$.
- The $\sin(\pi \cdot s)$ profile ensures zero velocity at the start and end of swing — smooth transitions.

6.4 Observed Results

Observation	LF	RF	LH	RH
Hip min angle (°)	-10.00	-10.00	-10.00	-10.00
Hip max angle (°)	+10.00	+10.00	+10.00	+10.00
Knee min angle (°)	25.00	25.00	25.00	25.00
Knee max angle (°)	75.00	75.00	75.00	75.00
Phase offset (s)	0.000	0.500	0.250	0.750

All four legs produce identical angle profiles, shifted in time by their respective phase offsets. This confirms correct implementation of the trot gait pattern.

7. Question 6 — Motion Validation via Simulation

7.1 Simulation Approach

Two complementary simulation approaches were used:

- **MATLAB 3D Stick-Figure Animation:** A programmatic MATLAB simulation using forward kinematics to animate the leg stick figures in 3D (file: Q6_Pure_3D_Animation.m). This requires no additional toolboxes and runs on base MATLAB or GNU Octave.
- **MATLAB 9-Panel Validation Plot:** A comprehensive validation figure (Q6_Motion_Validation_Plots.m) showing hip/knee trajectories, foot paths, gait contact diagram, stability analysis, and 3D foot trajectories in one figure.
- **SimScale Cloud Simulation (Optional):** The STEP assembly file was uploaded to SimScale (simscale.com, free tier) for a rigid-body motion study, providing 3D browser-based animation with collision detection.

7.2 Forward Kinematics — Foot Trajectory Computation

The foot endpoint position in 3D space is computed at each time step using forward kinematics. With hip at the origin, x = forward, y = lateral, z = upward (negative = down):

Quantity	Formula	Meaning
x_knee	$L1 \times \sin(\theta_{\text{hip}})$	Knee X position (forward offset from hip)
z_knee	$-L1 \times \cos(\theta_{\text{hip}})$	Knee Z position (downward)
x_foot	$x_{\text{knee}} + L2 \times \sin(\theta_{\text{hip}} + \theta_{\text{knee}})$	Foot X position
z_foot	$z_{\text{knee}} - L2 \times \cos(\theta_{\text{hip}} + \theta_{\text{knee}})$	Foot Z position
y_foot	constant (= hip Y)	No lateral motion in 2-DOF leg

7.3 Computed Foot Trajectory Results

Metric	Value	Significance
Ground level (y)	$L1 + L2 = 200$ mm below hip	Robot stands 200 mm tall
Maximum foot lift (at mid-swing)	92.8 mm above ground	Sufficient clearance for obstacles
Foot forward stride range (X)	108.5 mm	Stride length in body frame
Foot X range	[8.5, 117.0] mm	Forward excursion during swing
Foot Y range	[107.2, 200.0] mm	Vertical range (107 = peak lift, 200 = ground)

7.4 Collision-Free Verification

- **Foot trajectory:** The foot follows a smooth elliptical arc — lifts off the ground, peaks at 92.8 mm clearance, and lands smoothly. No sudden discontinuities or jerks.

- Inter-leg clearance: Legs are attached at body corners separated by 190 mm (length) and 90 mm (width). Foot excursion in X is 108.5 mm, well within the per-leg allocated space.
- Body clearance: At maximum hip angle (10°) and maximum knee angle (75°), the thigh and shin remain below the body underside. No collision detected.
- Joint velocity: Computed as $d\theta/dt$ — both hip and knee velocities are continuous, confirming smooth motion (no impulse-like velocity spikes).

7.5 Validation Summary

Validation Check	Method	Result
Smooth foot trajectory	MATLAB FK + plot	PASS — elliptical arc, no discontinuities
No ground penetration	MATLAB FK check	PASS — foot $y \leq 200$ mm at all times
No body-leg collision	Geometric check	PASS — clearances verified analytically
Continuous joint velocities	MATLAB diff(angles)	PASS — no impulse spikes in $d\theta/dt$
Phase-correct foot timing	Contact diagram	PASS — diagonal pairs synchronised
Symmetric leg motion	Overlay all 4 legs	PASS — identical profiles, shifted in phase

8. Question 7 — Stability Check

8.1 Stability Analysis Method

Static stability for legged robots is assessed using the support polygon method. A robot is statically stable if and only if the vertical projection of its Centre of Mass (CoM) onto the ground plane falls strictly inside the convex hull formed by all feet currently in contact with the ground.

The stability margin is quantified as the shortest perpendicular distance from the CoM projection to the nearest edge of the support polygon — a larger margin indicates greater tolerance to external perturbations.

8.2 Robot Parameters for Stability Analysis

Parameter	Value	Notes
Body length (X)	300 mm	Hip-to-hip distance front-to-back (using 300 mm for stability calc)
Body width (Y)	150 mm	Hip-to-hip distance left-to-right (using 150 mm for stability calc)
Foot positions (body frame)	See table below	Symmetric at body corners
CoM position	(0, 0) mm	Assumed at geometric body centre
Foot ground contact	Point contact at foot ends	Each foot treated as a point on the ground

Leg	Foot X (mm)	Foot Y (mm)	Notes
LF — Left Front	+150	+75	Front-left corner
RF — Right Front	+150	-75	Front-right corner
LH — Left Hind	-150	+75	Rear-left corner
RH — Right Hind	-150	-75	Rear-right corner

8.3 Stability Analysis — Three Cases

Case 1: Trot Gait — Phase A (LF + RH in contact)

When the LF and RH diagonal pair is in stance, only two feet are in contact with the ground at positions (+150, +75) and (-150, -75).

Item	Value
Contact feet	LF (+150, +75) and RH (-150, -75)
Support shape	Line segment (diagonal)
Support polygon area	0 mm ² (degenerate — a line, not a polygon)
CoM on support line?	YES — CoM at (0,0) lies exactly on the diagonal LF-RH line
Stability status	MARGINAL — CoM on boundary, not inside polygon

Stability margin	0 mm
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Case 2: Trot Gait — Phase B (RF + LH in contact)

When the RF and LH diagonal pair is in stance, contact feet are at (+150, -75) and (-150, +75).

Item	Value
Contact feet	RF (+150, -75) and LH (-150, +75)
Support shape	Line segment (opposite diagonal)
CoM on support line?	YES — CoM at (0,0) lies exactly on the RF-LH diagonal
Stability status	MARGINAL — CoM on boundary
Stability margin	0 mm

Case 3: Static Stand — All 4 Feet in Contact

When the robot is standing stationary with all four feet on the ground, the support polygon is the rectangle formed by all four foot contact points.

Item	Value
Contact feet	LF, RF, LH, RH — all four
Support shape	Rectangle 300 × 150 mm
Support polygon area	$300 \times 150 = 45,000 \text{ mm}^2 = 0.045 \text{ m}^2$
CoM at (0,0)?	YES — exactly at rectangle centre
Distance to front/rear edges	150 mm
Distance to left/right edges	75 mm
Minimum stability margin	$\min(150, 75) = 75 \text{ mm}$
Stability status	STABLE — CoM well inside polygon

8.4 Stability Summary Table

Gait Phase	Feet in Contact	Support Shape	CoM Inside?	Stability Margin	Status
Trot — Phase A	LF + RH	Diagonal line	On boundary	0 mm	MARGINAL
Trot — Phase B	RF + LH	Diagonal line	On boundary	0 mm	MARGINAL
Static stand	All 4	300×150 mm rect.	Yes (centre)	75 mm	STABLE

8.5 Discussion — Trot Gait Stability

The marginal stability during trot gait phases is not a design flaw — it is an inherent characteristic of trot locomotion. For the trot gait to achieve zero stability margin, the CoM must lie exactly on the diagonal support line, which is only possible if the robot is perfectly symmetric and mass is uniformly distributed. In practice:

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- **Dynamic stability:** The trot gait is dynamically stable. The robot maintains balance through momentum and rapid leg exchange, not through static support polygon geometry.
 - **CoM forward bias:** Shifting the CoM 5–10 mm forward of centre would place it slightly ahead of the LF-RH diagonal, giving a small positive stability margin during Phase A stance.
 - **Stability at transition:** As one diagonal pair lands while the other lifts, there is a brief moment with all 4 feet in contact, providing full stability margin.
 - **Slow crawl gait:** For fully static stability throughout the cycle, a crawl gait (3 feet always in contact) would be required — but this sacrifices walking speed.

9. Question 8 — Design Limitations

9.1 Limitation 1: Restricted Workspace (2-DOF per Leg)

Description

Each leg has only 2 degrees of freedom (hip flexion/extension and knee flexion/extension), both rotating in the sagittal plane. This means the leg can only move forward and backward in a single vertical plane aligned with the walking direction.

Consequences

- No lateral (side-to-side) motion: The robot cannot step sideways, turn with a small radius, or compensate for lateral perturbations such as a sideways push.
- No terrain adaptation: On uneven ground, the robot cannot abduct or adduct individual legs to maintain footing. All feet must contact at the same height.
- No turning capability: Without a third DOF (hip abduction/adduction or body yaw), the only way to turn is by differential speed between left and right legs, which is difficult with only sagittal-plane joints.
- Reduced reachable workspace: The 2-DOF workspace is a 2D arc, not a 3D sphere. A 3-DOF leg would add $\pm 20^\circ$ of lateral reach, roughly tripling the accessible ground area per leg.

Proposed Fix

Adding a third hip joint rotating about the Z-axis (vertical axis) for abduction/adduction would expand the workspace to 3D. This is implemented in real quadruped robots such as Boston Dynamics Spot and MIT Cheetah, which use 3-DOF legs.

9.2 Limitation 2: Marginal Trot Stability

Description

As demonstrated in Question 7, trot gait places only 2 feet in contact with the ground at a time, and the support structure is a line (not a polygon). The stability margin is exactly 0 mm, meaning the CoM sits precisely on the support edge.

Consequences

- Any CoM offset (uneven mass distribution, payload) will push the CoM off the support line, causing tipping in one direction.
- External perturbations (wind, surface irregularities) cannot be resisted statically — the robot depends entirely on dynamic balance.
- The robot cannot carry asymmetric payloads (e.g., a sensor mounted to one side) without additional balance compensation.
- At low walking speeds where dynamic effects are small, the robot may become unstable and fall.

Proposed Fix

Use a crawl (walk) gait instead of trot for low-speed operation. In a crawl gait, 3 of 4 feet are always in contact, maintaining a triangular support polygon with a non-zero stability margin. The trade-off is lower walking speed.

9.3 Limitation 3: Rigid Body Assumption (No Compliance)

Description

The simulation models all links and joints as perfectly rigid bodies with no elasticity, compliance, or damping. In reality, leg impacts with the ground generate high-frequency impulse forces.

- Impact loads: Without springs or dampers, foot impact forces are transmitted directly to the servo motors and structural links, risking mechanical failure.
- No terrain sensing: Rigid-body joints cannot sense or adapt to ground irregularities. Pre-planned joint trajectories (open-loop control) will cause stumbling on uneven terrain.
- Servo stress: At 1 Hz gait frequency with 100 mm link length, foot velocities can reach 0.5 m/s. Rigid joint stops generate high torque spikes.

9.4 Limitation 4: Pre-Planned Trajectory (No Feedback Control)

Description

The joint angles are computed as open-loop sinusoidal functions of time with no sensor feedback. There is no control loop that corrects for disturbances or joint position errors.

- If a leg slips or a foot contact is delayed, the pre-planned timing will be wrong and the gait will break down.
- No closed-loop joint control means accumulated errors are not corrected — the robot will drift from its intended path.
- Real quadruped robots use inertial measurement units (IMU), foot contact sensors, and joint encoders with closed-loop PID or model-predictive control.

9.5 Design Limitations Summary

#	Limitation	Impact	Proposed Solution
1	2-DOF leg — no lateral motion	Cannot turn or step sideways	Add 3rd hip DOF (abduction/adduction)
2	Trot — marginal stability	CoM on support boundary — no perturbation tolerance	Use crawl gait at low speed OR shift CoM forward
3	Rigid body — no compliance	High impact loads, no terrain adaptation	Add passive springs to shin; soft foot pads
4	Open-loop trajectory	No disturbance rejection	Add IMU + joint encoders + PID closed-loop control

10. Conclusion

This project successfully designed and analysed a simplified quadruped robot, addressing all eight questions of the assignment brief. The key outcomes are summarised below:

Question	Deliverable	Status
Q1 — Mechanical Design	200×100×40 mm body, 4× 2-DOF legs, L1=L2=100 mm	Complete
Q2 — CAD Modelling	FreeCAD assembly with correct symmetry, exported as STEP	Complete
Q3 — Joint Limits	Hip: $\pm 30^\circ$, Knee: 10° – 90° , both Y-axis revolute joints	Complete
Q4 — Gait Calculation	$T=1.0s$, $\beta=0.60$, trot phase offsets, IK/FK verified	Complete
Q5 — MATLAB Trajectories	Hip $\pm 10^\circ$, Knee 25° – 75° , all 4 legs plotted, correct phases	Complete
Q6 — Motion Validation	MATLAB 3D animation + 9-panel validation, smooth motion verified	Complete
Q7 — Stability Check	4-foot: 75 mm margin (STABLE); trot: 0 mm (MARGINAL)	Complete
Q8 — Design Limitations	4 limitations identified and solutions proposed	Complete

The project demonstrates that a simplified quadruped can be successfully designed using basic kinematic principles. The forward kinematics model accurately predicts foot trajectories, and the trot gait produces smooth, continuous joint motion. The identified limitations — restricted 2-DOF workspace, marginal trot stability, rigid body assumption, and open-loop control — are well-understood engineering trade-offs that form the basis for real-world quadruped robot design improvements.

The tools used (FreeCAD, MATLAB, Python) are accessible and effective for educational robotics projects. The report and associated code files are ready for submission as specified in the assignment guidelines.

— End of Report —